

GENERAL SPECIFICATION G4.73

Odour Containment Covers

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G4.73 ODOUR CONTAINMENT COVERS

G4.73.1 General

- (a) This specification covers the general requirements concerning the performance, design, supply, installation supervision, testing, and commissioning of odour covers and all related appurtenances.
- (b) Odour containment covers must satisfy relevant statutory regulations and the latest edition of all relevant international, harmonised European and British Standards and Singapore Standards and Code of Practices.
- (c) The function of the covers is to contain odorous air in the headspaces above top water level in the units. The odorous air will be extracted and treated by the odour control system.
- (d) The works shall be supplied complete with all minor and incidental items necessary for the proper functioning of the whole system, even though they may not be specifically detailed or mentioned.
- (e) This specification shall be read in conjunction with:
 - (i) The relevant Mechanical discipline related Particular Specification;
 - (ii) The specific Technical Data Schedules attached as part of the Mechanical Particular Specifications (if applicable);
 - (iii) The relevant Electrical General and Particular Specifications; and
 - (iv) The relevant Instrumentation and Control and Automation (ICA) General Specification G6 and Particular Specification P6.

G4.73.2 Standards and Reference Documents

- (a) Equipment and components shall be in accordance with the latest version of applicable codes and regulations, including the following:
 - (i) Singapore Standards and Codes of Practice;
 - (ii) British Standards (BSI);
 - (iii) European Standards (EN);
 - (iv) International Standards Organisation (ISO) Standards;
 - (v) Statutory regulations and codes;
 - (vi) OH&S requirements; and
 - (vii) The Board's specifications;
- (b) This specification shall be read in conjunction with the following Standards, which are deemed to form a part of this specification. Reference to a Standard shall mean the Standard current as at the time of providing the works. All materials, equipment and workmanship shall comply with these Standards or other industry recognised standards in the absence of specified standards, unless expressly noted otherwise. Where requirements in the specifications and Standards are inconsistent or conflicting, the more stringent requirement shall take precedence.

Standard	Subject
BS2782-10 Method:1001	Methods of testing plastics. Glass reinforced plastics. Measurement of hardness by means of a Barcol impressor.
ASTM E-84/ BS 476-7	Fire tests on building materials and structures.
BS 903-A18	Physical testing of rubber. Determination of equilibrium water vapour absorption.
BS ISO 812:Part 114	Rubber, vulcanized or thermoplastic. Determination of low-temperature brittleness.
ASTM D4097-88	Standard Specification for Contact Molded Glass Fiber Reinforced Thermoset Resin Chemical Resistant Tanks.
ASTM D635	Standard Test Method for Rate of Burning and/or Extent and Time of Burning of Plastics in a Horizontal Position.
ASTM E-84	Standard Test Method for Surface Burning Characteristics of Building Materials; Class 1 Rated.

G4.73.3 Definitions and Abbreviations

- (a) For the purposes of this specification the following definitions shall apply:
- (i) The following is a list of abbreviations and definitions of units which may be used in this specification:

Abbreviations/ Definitions	Subject
C	Celsius
dB(A)	Decibel, A scale
DMS	Dimethyl sulphide
DMDS	Dimethyl disulphide
DN	Nominal Diameter
EPDM	Ethylene Propylene Diene Monomer
FRP	Fiberglass Reinforced Plastic
HDPE	High Density Polyethylene
H ₂ S	Hydrogen sulphide
Am ³ /hr	Actual Cubic metres per hour, i.e. gas flow at actual gas temperature and pressure
Nm ³ /hr	Normal cubic metres per hour, i.e. at 20 °C, 1 bar, and 36% RH
OCS	Odour Control System
OH&S	Occupational Health and Safety

Abbreviations/ Definitions	Subject
Ou	Odour unit as defined in AS/NZS 4323.3
P.E.	Professional Engineer
PUB	Public Utilities Board
PVC	Polyvinyl Chloride
P&ID	Process and Instrumentation Diagram
ppm	Parts Per Million
ppm _v	Parts Per Million (volume basis)
Ppb	Parts Per Billion
ppb _v	Parts Per Billion (volume basis)
Pa	Pascals
kPa	Kilopascals
RH%	Relative Humidity
RSC	Reduced Sulphur Compound
SOP	Standard Operating Procedures
VOC	Volatile Organic Compound
WLL	Working Load Limit

G4.73.4 Submittals

- (a) The Contractor shall provide complete fabrication and assembly drawings, together with detailed specifications and datasheets covering materials, completed assemblies, and accessories forming the supplied odour containment covers. The datasheets and specifications for each component of the odour containment covers shall include, but shall not be limited to, the following:
 - (i) Details of any non-conformances relating to this technical specification;
 - (ii) Manufacturer's catalogue information, technical brochures and Operations and Maintenance manuals on the proposed cover type;
 - (iii) Fully completed Technical Data Schedules for the odour containment covers as attached in the Appendices to the Mechanical Particular Specifications;
 - (iv) Clearly defined scope of supply and installation, including any assumptions;
 - (v) Basic description of the odour containment cover system offered;
 - (vi) Preliminary odour cover structural loading data including dead loads, live loads, wind loads, earthquake loads, and uplift loads, applied to concrete structures and all connecting surfaces. Preliminary anchor details shall also be provided;

- (vii) Reference lists and relevant case studies demonstrating the Contractor's previous experience;
- (viii) A list of priced and un-priced spare parts for both commissioning (12 months) and 12-month operation; and
- (ix) List of any special tools and lifting devices.
- (x) Drawings, technical schedules, lists, and submittals, with the following formats or contents as a minimum, but not limited to:
 - 1) 2D or 3D files in AutoCAD (DWG) or Microstation (DGN) format of complete odour cover assemblies and dimensions, including number and size of cover panels, number, size, and location of hatches, number, size, and location of nozzles, lifting lugs, opening handles and all other electrical, mechanical and instrumentation interface points;
 - 2) General layout drawings (with plan, elevations, sections and details) showing the cover assemblies and locations of interface connections with Work by Others;
 - 3) Drawings shall be produced in accordance with the Board's standard drawing symbols and templates;
 - 4) Final odour cover structural loading data including dead loads, live loads, wind loads, earthquake loads, and uplift loads, applied to concrete structures and all connecting surfaces. Provide final anchor details. Provide template for cast-in-place anchors. Provide summary of calculated deflections for cover components as stated herein;
 - 5) Dimensioned outline drawings for the civil structures (to be supplied by others) required to support and provide access to the equipment. These drawings shall show in sufficient detail all tolerances, special requirements, details of conduits required for equipment and instrumentation, finishes and equipment loadings to allow others to undertake the structural detailed design. Civil and structural design of covers shall be carried out by a qualified P.E. and the design shall be endorsed by a P.E. (Structural) Singapore and submitted to S.O.;
 - 6) Completed and finalised Technical Schedules attached to the Appendices of the relevant Particular Mechanical Specifications;
 - 7) Equipment lists;
 - 8) Installation and commissioning instructions, including installation checks, pre-commissioning checks, commissioning, and on-site testing;
 - 9) Recommended spare parts lists for equipment items which are subject to wear or consumption;
 - 10) List of proposed suppliers and subcontractors;
 - 11) Delivery schedule;
 - 12) Statement of warranty;
 - 13) Design Certificate by P.E.;

- 14) Installation and commissioning instructions, including installation checks and on-site testing;
 - 15) Shop and field painting systems;
 - 16) Inspection and Test Plans;
 - 17) Packaging and transport details;
 - 18) Recommended spare parts lists. The spare parts shall include gaskets, etc., which are subject to wear; and
 - 19) A Design Schedule detailing the list of Design Deliverables, Drawings, Calculations, and Specifications. The schedule will also contain the list of deliverables and respective dates of completion.
- (b) No construction will be permitted until designs have been reviewed and approved by the S.O.. Review and approval of designs by the S.O. does not relieve the Contractor of his responsibility in meeting all the requirements in this Specification.
- (c) The drawings shall include the minimum clearance distance required for the removal of plant and equipment and its associated components with the equipment in situ and termination points for the connection of external services.

G4.73.5 Manufacturer's Qualifications

- (a) The materials covered by these specifications are intended to be standard equipment of proven reliability and as manufactured by a reputable manufacturer having experience in the production of such equipment. The equipment furnished shall be designed, constructed, and installed in accordance with the best practices and methods and shall operate satisfactorily when installed as shown in the Drawings and operated per Manufacturer's recommendations.
- (b) The Contractor shall provide a list of at least THREE (3) installations for which the Manufacturer have been responsible, each associated with the containment of air contaminated with H₂S and other typical organic municipal/industrial wastewater odours. The list shall include contact names, telephone numbers, capacity, and length of operation for each installation. Installations shall be of comparable size to the type specified for this project, either within or outside Singapore.
- (c) The Manufacturer shall be recognised in the design, manufacture, supply, and installation of odour covers and shall have at least 5 years' experience in the design, fabrication, supply, and installation of odour covers similar to the type specified for this project.
- (d) The Manufacturer shall submit an installation list with contact information for the equipment quoted verifying to the S.O. that the specified Manufacturer's qualifications are met.

G4.73.6 Odour Cover Types

- (a) General

- (i) The odour cover types described in this section are acceptable to the Board. Contractors must only propose the odour cover types as outlined in this document and approved by S.O.; and
 - (ii) All covers proposed, including supporting equipment, components, and accessories offered shall be of an acceptable technology.
- (b) Anodised Aluminium “Non-Trafficable” Dome covers:
 - (i) Dome Framing System: Fully triangulated anodised marine grade aluminium space truss with anodised marine grade aluminium panels. The completed dome shall be clear-span, low-rise structure with maximum height of dome as shown on submitted Drawings by the Contractor;
 - (ii) Cover system shall be designed to be self-supporting from the tank wall. Panel nominal thickness shall be minimum 1.3 mm;
 - (iii) Continuous gasket seals from either Neoprene or Ethylene Propylene Diene Monomer (EPDM) shall be affixed to the perimeter of the panels and to the total cover to ensure a relatively airtight system; and
 - (iv) The cover system must be sufficiently air-tight such that gas from the tank contents can be withdrawn in a controlled manner by an exhaust system creating a normal operating vacuum under the covers of between 0 and - 10 mm water gauge (under negative pressure).
- (c) Anodised Aluminium “Trafficable” Panel covers:
 - (i) Covers shall be profiled anodised marine grade aluminium, clear-span, and self-supporting from supporting walls. Covers shall be assumed to be semi-permanent and stackable for occasional maintenance;
 - (ii) Cover system shall be designed to be self-supporting from the tank wall. Panel nominal thickness shall be minimum 1.3 mm;
 - (iii) Continuous gasket seals from either Neoprene or EPDM shall be affixed to the perimeter of the panels and to the total cover to ensure a relatively airtight system; and
 - (iv) The cover system must be sufficiently air-tight such that gas from the tank contents can be withdrawn in a controlled manner by an exhaust system creating a normal operating vacuum under the covers of between 0 and - 10 mm water gauge (under negative pressure).
- (d) FRP Domed “Non-Trafficable” Covers (including sliding type domed non-trafficable covers):
 - (i) Covers shall be monolithic construction profiled FRP, clear-span and self-supporting from supporting walls;
 - (ii) Covers shall be assumed to be semi-permanent and stackable for occasional maintenance;
 - (iii) Covers shall be bolted together to achieve an airtight gas seal;
 - (iv) Self-adhesive 30 mm x 6 mm continuous soft neoprene foam gasket seals shall be affixed to the perimeter of the cover;

- (v) The cover system must be sufficiently air-tight such that gas from the tank contents can be withdrawn in a controlled manner by an exhaust system creating a normal operating vacuum under the covers of between 0 and - 10 mm water gauge (under negative pressure); and
- (vi) Sliding type covers shall satisfy the following requirements in addition to the above:
 - 1) Sliding type covers shall have a smooth and flush exterior and interior surface without any projected reinforcing rib to affect its sliding and interlocking functions;
 - 2) Sliding type covers shall be designed to form a continuous solid surface which shall be airtight by sealing with sealed neoprene foam gasket seals;
 - 3) Movable cover sections shall be mounted along chrome plated heavy duty industrial grade linear bearing rods with Grade 316 Stainless Steel bearing blocks for smooth action and durability. The covers shall not be subjected to derailment or misalignment of the rollers from the sliding tracks due to thermal, wind, static or dynamic loads while it is in a stable or mobile condition; and
 - 4) Each movable cover section shall have good interlocked and double sealed mechanical joints with neoprene or EPDM gaskets to minimise odorous air leakage. Grade 316 Stainless Steel mechanical locks shall be provided fix the position of each movable cover.
- (e) FRP sandwich polyurethane “Trafficable” panels:
 - (i) Covers shall be profiled FRP, clear-span, and self-supporting from supporting walls;
 - (ii) Covers shall be suitable for standard duty, with a panel nominal minimum thickness of 76 mm;
 - (iii) Covers shall be assumed to be semi-permanent and stackable for occasional maintenance;
 - (iv) Covers shall be bolted together to achieve an airtight gas seal;
 - (v) Self-adhesive 30 mm x 6 mm continuous soft neoprene foam gasket seals shall be affixed to the perimeter of the cover; and
 - (vi) The cover system must be sufficiently air-tight such that gas from the tank contents can be withdrawn in a controlled manner by an exhaust system creating a normal operating vacuum under the covers of between 0 and - 10 mm water gauge (under negative pressure).
- (f) Structurally supported fabric (geotextile) cover system:
 - (i) Covers shall be of low profile construction that does not interfere with internal tank components. Covers and support systems shall be assumed to be semi-permanent and retractable for access to the internal tank components;

- (ii) Retractable cover panels shall be battened at the tank perimeter on one side and held in position by detachable connections on the other three sides;
- (iii) Support arches shall be provided at intermittent locations along the tank to support the fabric cover panels. Arches must be securely fastened to the tank walls. Where required, walkways may contain hatches for access to tank components and liquid sampling;
- (iv) The retractable cover system must be sufficiently air-tight such that gas from the tank contents can be withdrawn in a controlled manner by an exhaust system creating a normal operating vacuum under the covers of between 0 and -10 mm water gauge (under negative pressure). Air-tight connections for off-gas removal pipes must also be fitted on the retractable covers; and
- (v) The cover system, including fabric, arches and any required walkways must be designed to support loads which are specified in the General Specifications for Civil and Building Works (G3.1 to G3.44) and relevant to site specific conditions.

G4.73.7 System Guarantees

- (a) The Contractor shall warrant that all equipment provided by the Contractor, as part of the odour containment covers, shall perform to the requirements as outlined in this General Specification and the Particular Specifications.
- (b) The covers shall maintain airtightness and loading capabilities specified with minimal maintenance.
- (c) The Contractor shall provide details of maintenance procedures and frequency, which are required to ensure these basic performance characteristics are sustained, in the Contractors proposal.
- (d) The Contractors shall provide a written undertaking, that the Contractor will be responsible for all costs related to maintenance (more than routine maintenance described above and detailed in the Operations and Maintenance Manual), repair or replacement, in whole or in part, of the cover to enable it to continue meeting the performance criteria described in this document for a period of ten (10) years after installation.
- (e) Exercise of this warranty will not be limited by any condition other than accidental or intentional misuse of, or damage to, the cover caused by the Board, its agents or successors.

G4.73.8 General Design Requirements

- (a) General
 - (i) The odour control system will extract air from beneath the odour containment covers by means of connected ductwork and a fan arrangement, which will deliver the odorous air to the odour treatment facility.

- (ii) To ensure adequate ventilation and prevent fugitive odours, the space under the covers shall be maintained under a nominal negative pressure. The required negative pressures are outlined in the table below.

Table G4.73-1: Design Requirements

Design Requirements	Requirements
Operating pressure	50 Pa.g negative
Design pressure	-250 Pa.g to + 250Pa.g
Design life	20 years

- (iii) The cover system shall remain sufficiently airtight to allow this negative pressure to be sustained under all the service conditions specified herein. Gaps between covers and joints shall be sealed with appropriate gaskets (Neoprene or EPDM) or overlap systems. Gaps shall not exceed 1 mm in width.
 - (iv) The design shall include all fittings and appurtenances required for the successful operation of the covers. The cover system shall include all supporting structural members.
 - (v) The headspace below the covers will contain a corrosive atmosphere and must be fully lined. Horizontal flanges or other structural surfaces which may collect condensation below the covers shall be avoided. Where horizontal structural surfaces cannot be avoided, suitable provision shall be made to avoid pooling, and for drainage of condensation.
 - (vi) Covers and appurtenances shall have a design life of at least 20 years.
- (b) Odour Covers
- (i) The covers shall incorporate flanged nozzles for the air inlet assemblies as shown on referenced drawings in the Particular Specifications. The air intake flanges shall be located a minimum of 150 mm above the cover surface to allow for ease of assembly/disassembly.
 - (ii) The cover shall be located above the top of tank and beneath any access bridges.
 - (iii) The covers, fixings, and associated operating equipment shall be located wholly above the maximum top water level for all flow conditions.
 - (iv) The covers shall be fixed to the concrete tank copings.
 - (v) The completed work shall include all permanent safety devices necessary for safe access, operation and maintenance and to meet relevant safety regulations. This will include handrails and signs around all fail safe opening panels
 - (vi) The cover shall be made up of removable and re-sealable sections, or shall be designed with suitable man access points. Sealable access hatches shall be provided for:
 - 1) Access to all isolation penstocks and stop boards to facilitate actuation of penstocks and stop boards;

- 2) Access to mechanical equipment, including but not limited to screens, aeration headers, mixers, sludge scrapers, and chemical dosing points, to facilitate inspection, cleaning, and maintenance if required; and
 - 3) Access to effluent launders (multiple locations), scum collectors, and hoppers to facilitate inspection and cleaning.
- (vii) The cover shall be suitably designed to completely cover the air space between the top of the liquid surface and the top of the enclosed walls, and contain any odorous air within.
- (viii) The design shall include all fittings required for the successful operation of the covers.
- (ix) The covers shall be designed to accommodate all the existing pipework and connections to mechanical, electrical, and instrumentation items located within the covered area.
- (c) Odour Cover Seals and Hold Down Bolts
 - (i) The cover system shall remain sufficiently airtight to allow the required negative pressure to be sustained under all service conditions. Adequate sealing between any centre drive mechanisms/mixers, including any process pipes and the odour cover shall be provided to prevent fugitive foul air from migrating to the motor drive area.
 - (ii) Gaps between covers and joints shall be sealed with appropriate gasket of Neoprene or EPDM, sealant, or overlap systems. Gaps shall not exceed 1 mm in width.
 - (iii) Sealants/gaskets shall be provided as required around covers and inspection hatches to prevent the escape of odours when closed.
 - (iv) The edge of the covers shall be fixed to the top of concrete walls. Any bearing plates required to fix the covers to the tank walls shall be of suitable material for the environment. Covers shall be designed to ensure that it does not cause a tripping hazard.
 - (v) Hold down bolts shall be Grade 316 stainless steel, with Grade 316 nuts and washers, unless the covers are of a material that a galvanic reaction may occur.
- (d) Lifting Provisions and Safety considerations
 - (i) The completed work shall include all permanent safety devices necessary for safe access, operation and maintenance and to meet relevant safety regulations.
 - (ii) Lifting provisions shall be provided on all covers requiring mechanical lifting assistance. Such provisions may take the form of lugs, eye bolts or slinging cradles. All lifting lugs are to be certified and stamped with their Working Load Limit (WLL).
 - (iii) The design must be sufficiently robust to withstand an appropriate number of removal and replacement operations during the life of the covers, depending on their access type designation and function.

- (iv) Lifting provisions shall be provided on all covers indicated to be removable. For removable covers, panels shall be easy to remove and the lifting force shall not exceed the dead weight of the module.
 - 1) The dead weight of the cover panel shall not exceed 25 kg/m². Lifting provisions shall take the form of handles concealed or flush to surface; and
 - 2) The design must be sufficiently robust to withstand a number of removal and replacement operations during the life of the covers.
- (v) Sliding type covers, if specified, shall ensure that each movable section will be able to slide along the tracks when manned by one person on a single side. The maximum allowable sliding force (pushing and pulling) at a single side of the cover section shall not be more than 250 N.
- (e) Connections on Covers
 - (i) The inlet and outlet connections on covers shall be located on opposite sides of the process unit or Installation being covered, to prevent short circuiting of air flows.
 - (ii) Outlet connections on covers shall be positioned as far as practical from access hatches if provided.
 - (iii) Where a number of different items of covered process equipment are connected in series, access hatches shall be designed so that they cannot be inadvertently left open, as this would disrupt the air flow through the entire installation.
- (f) Materials of Construction
 - (i) All covers, fixings, hinges, handles and support beams supplied under this work must be suitable for the environmental conditions detailed in the Particular Specifications. Materials shall be suitable for contact with potentially corrosive fluids and environment contained under the covers as outlined in the Particular Specifications.
 - (ii) It is expected that anodised marine grade aluminium alloys, chemical grade isophthalic polyester or polyurethane based FRP, or reinforced polyurethane coated PVC or reinforced polypropylene membrane materials will be required for the odour covers as specified. The Contractor may offer alternative materials but must give examples of where the alternative materials have been used successfully in similar applications.
 - (iii) The following materials are acceptable for the cover systems:
 - 1) Anodised aluminium alloys 5052-H36 or 5086-H34 (formed aluminium sheet) or better and 6061-T6 or 3003-H16 (extrusions) or better;
 - 2) Profiled FRP covers appropriately designed for non-trafficable areas. Material shall be chemical grade isophthalic polyester resin with 'E' glass reinforcement and a 'C' glass surface tissue for the internal corrosion barrier. Aperture Nexus 1012 shall be used for abrasion resistance. Coloured UV resistant coating on external (exposed) surface. Colour as approved by the S.O.; and

- 3) FRP sandwich polyurethane panels appropriately designed for trafficable areas.
- (iv) Full specifications and details of the materials proposed for the cover system shall be provided with the Contractor's submission.
- (v) Any dissimilar materials shall be separated from each other using approved gaskets or coatings. Sealants/gaskets will be provided as required around covers and inspection hatches to prevent the escape of odours when closed.
- (g) Protective Coatings
 - (i) Materials of construction shall be suitable for the application and/or be provided with adequate corrosion protection.
 - (ii) Details of the materials and corrosion protection coatings of components of the cover assemblies are to be provided by the Contractor with the tender submission.

G4.73.9 Mechanical Design Requirements

- (a) Rainwater Disposal
 - (i) Unless specifically noted otherwise, rainwater falling on the covers shall be collected and discharged through the covers into the covered tanks so as to:
 - 1) Prevent flooding of pedestrian access on the covers;
 - 2) Prevent discharge off the covers onto surrounding areas; and
 - 3) Prevent damage to the covers.
 - (ii) Rainwater disposal requirements shall be met for a design storm of 1 in 5 years with a 5-minute duration (190 mm/hr).
 - (iii) All drain piping shall extend a minimum of 150 mm below the bottom water level or be provided with appropriate trap seal to prevent escape of odours out the drain system. Any drain piping proposed to be extended into the process fluid shall require the S.O.'s approval.
- (b) Access and Service Requirements
 - (i) The Particular Specifications indicate two types of covered areas, designated as "operator accessible" and "maintenance accessible".
 - 1) "Operator accessible" areas surround or cover equipment or areas of the process units that will be accessed daily or weekly to allow operators to observe the process, operate stop boards or equipment, or to carry out routine cleaning or sampling.
 - Where requirements for access are indicated, hatches shall be provided of a size and weight that can be handled by one person;
 - If the hatches are loose, specific provisions shall be made for stacking the hatch securely while the opening is in use. If the hatches are hinged, the hinge system shall be sufficiently robust to withstand frequent use over the design life;

- Hinged hatches shall either fold back flush with adjacent covers, or be provided with a latchable stay to hold the hatch securely in the open position;
 - For “operator accessible” hatches, if the size of the hatch opening is sufficiently large to permit a person to fall through the opening, removable guard rails, or a similar system, shall be provided and stored adjacent to the hatch to enable the operators to barrier the opening. Easily accessible anchorages for safety lines (attachable by carabineer) may be considered as an alternative to guard rails, depending on the location;
 - Some “operator accessible” areas surround stop boards or stop log grooves required for isolating sections of the process units. Stop boards may protrude through the covers when in the open position. Suitable sealing strips shall be provided to the covers in this area to maintain an air seal between the cover and the stop board; and
 - All odour covers and hatches designated as “operator accessible” shall be sealed with gasket made of Neoprene.
- 2) “Maintenance accessible” areas are those where only occasional access will be required to permit maintenance or replacement of equipment located below the covers. The covers in these areas may be bolted in place. The size and weight of the covers shall be such as to permit removal by two persons. If larger or heavier covers are proposed, a lifting gantry shall be provided above the cover to facilitate removal.
- Where mechanical assistance is required for removal of maintenance access covers these and other covers affected by their removal and replacement shall be able to withstand the loading experienced. Lifting lugs must be certified and stamped with the WLL;
 - The cover system shall accommodate miscellaneous penetrations as shown on the drawings. Where the penetrations are designated as dosing points, a suitably sized tube shall protrude through the cover to below top water level, to maintain the air seal;
 - The covers shall afford access to all valves and equipment located under them even though not shown on the drawings; and
 - All odour covers and hatches designated as “maintenance accessible” shall be sealed with gasket made of Neoprene or EPDM.
- (ii) Odour containment covers may be “trafficable” or “non-trafficable” type as specified. Areas designated for “Trafficable” or “Non-Trafficable” type covers are outlined in the Particular Specifications.

- 1) “Non-trafficable” areas are defined as areas not subject to routine pedestrian access
 - Covers in these areas shall be designed for short-term trafficable loads to accommodate short duration live loads due to infrequent maintenance activities;
 - The covers shall be designed for the appropriate loading as specified in the Particular Specifications; and
 - All non- trafficable covers shall be enclosed by hand rails and kick plates.
 - 2) “Trafficable” areas / pedestrian access
 - Areas designated “trafficable” will be subject to pedestrian loading. The stiffness of all structural elements shall be such that deflections are within acceptable industry and relevant standards;
 - Pedestrian access shall be provided to the full extent of the odour ductwork and maintenance areas around all extraction points and control valves/dampers;
 - Deflections caused by people walking on the covers shall not cause discomfort to other persons on the covers nor prevent them from carrying out their work;
 - Pedestrian access will also be required to all operator access areas and to allow removal of all covers designated as maintenance access;
 - All trafficable covers shall be provided with a non-slip surface; and
 - Trip hazards shall be eliminated wherever possible. Where a trip hazard is unavoidable, warning signs shall be erected and the area shall be painted or otherwise coloured to provide clear visual indication of the hazard.
 - 3) “Lay-down” designations indicate areas where temporary equipment may be placed during operations or maintenance operations. The covers shall be designed for the appropriate live loading as specified under the design requirements.
- (iii) All covers shall be bolted in place to allow ease of removal for infrequent tank cleaning. The size and weight of cover panels shall be such as to permit removal by two persons. Any one removable cover panel shall not weigh more than 40 kg. If larger or heavier covers are proposed, a lifting gantry shall be provided above the covers to facilitate removal.
 - (iv) For removable cover panels, adjacent cover areas shall be designed to withstand the additional load of a minimum of four stacked panels.
 - (v) The cover system shall accommodate miscellaneous penetrations, pipe supports and the drive mechanism. The covers shall afford access to all valves and equipment located under them.

- (vi) Adequate labelling shall be used to distinguish between “trafficable” and “non-trafficable” covers. “Non-trafficable” covers shall be clearly labelled “Non- Trafficable”. The stiffness of all structural elements shall be such that deflections are within acceptable industry standards.
- (c) Hatches
 - (i) Hatches shall be provided of a size and weight that can be handled by one person (less than 20 kg).
 - (ii) Unless otherwise noted in the Particular Specifications, hatches shall be hinged. Hinge assemblies shall be sufficiently robust to withstand frequent use over the design life.
 - (iii) Hinged hatches shall either fold back flush with adjacent covers, or be provided with a latchable stay to hold the hatch securely in the open position.
 - (iv) Grab handles for hatches shall be retractable in to the hatch so as not to cause a trip hazard.
 - (v) Hatch design must ensure close alignment of cover parts so that the hatch seals correctly when closed.
 - (vi) Rectangular hatches should latch at each end so that regular opening and closing does not distort them. The design must ensure that the hatch seals correctly when closed.
 - (vii) Fall protection grating system shall be provided below the hatches and cover, The system shall be visible. Type 316L stainless steel hold open device shall be provided to lock the cover automatically in the open 90 degreed fully open position. Provision for locking to prevent unauthorized opening shall be included.
- (d) Anchor Bolts
 - (i) Anchor bolts for cover systems shall be sized, supplied and installed by the Manufacturer.
 - (ii) Sizing of all anchor bolts shall be undertaken by a P.E. of the relevant discipline employed by the Manufacturer.
 - (iii) The Contractor shall submit the following details to the S.O. as early as possible in the contract period to enable the design of support structure to be approved:
 - 1) Bolt specification, sizes and plan arrangement;
 - 2) Static and dynamic loads; and
 - 3) Embedment length into concrete (where applicable).
 - (iv) The anchor bolts shall be designed for the relevant combinations of all loads. If applicable the effects of fatigue shall be considered in the bolt design.

- (v) Unless noted otherwise in the Particular Specifications, anchor bolts shall be multifix type and manufactured from Grade 316 stainless steel. Neoprene material or similar protection shall be provided between dissimilar metals.
- (vi) Anchor bolts embedded in concrete shall be Grade 316 Stainless steel. The centreline of bolts shall be located at a minimum distance of 100 mm away from any concrete edge.
- (vii) In accordance with the project construction schedule, the equipment Manufacturer shall provide set-out templates for all anchor bolt group locations.
- (viii) The location of anchor bolts shall be in accordance with the tolerances nominated below.

Table G4.73-2: Anchor Bolt Locations Tolerances

Element/Dimension	Lower Permitted Deviation	Upper Permitted Deviation
Location of Centre of Gravity of group of bolts	-6 mm	+6 mm
Location of bolts relative to each other in one group	-3 mm	+3 mm

- (e) **Cover Installation Requirements**
 - (i) Where the covered areas terminate above open tanks, an appropriate air seal shall be provided between the underside of the cover and top water level. This seal shall not compromise the hydraulic performance of the plant by introducing a head loss of more than 5 mm. It is envisaged that the air seal may take the form of overlapping polythene or PVC strips hung vertically from the covers and extending to just below bottom water level;
 - (ii) Where indicated on the drawings, the cover system shall incorporate stainless steel spigots for connection to the odour extraction ductwork. The spigots shall have sufficient allowance to allow 2 stainless steel worm drive clamps to fasten a flexible hose to them;
 - (iii) The Contractor shall provide suitable equipment and trained personnel for assembly and installation of the covers;
 - (iv) Erection shall be supervised by the cover Manufacturer and performed by skilled and qualified personnel in accordance with the Manufacturer's standards;
 - (v) Erect plumb and level and in proper alignment;
 - (vi) Field cutting and patching will be permitted only with the S.O.'s approval;

- (vii) Sealant shall be Neoprene or silicone based (EPDM) and joints shall be tooled slightly concave after sealant is installed. Care shall be taken to keep sealant confined to the joint area in a neat manner. Any sealant applied outside the joint shall be removed so that panels will be free from misplaced sealant; and
- (viii) All gasket materials shall be continuous. Splices will not be allowed.
- (f) Mechanical Lifting Requirements
 - (i) Where mechanical assistance is required to remove and replace maintenance access covers the overall layout of covers shall provide corridors with suitable surface and load carrying capacity to allow access for such equipment from the road adjacent the structures to the required point of use.

G4.73.10 Structural Design Requirements

- (a) General
 - (i) The performance specifications provided below are for the information of the Contractor. The general loadings, load combinations and guidelines of features shall refer to the General Specifications for Civil and Building Works (G3.1 to G3.44), which the Contractor shall incorporate into the final design. Any changes to these features that the Contractor may find necessary in his design work must be approved by the S.O..
 - (ii) The cover system shall include all supporting structural members.
 - (iii) The headspace below the covers contains a corrosive atmosphere as outlined in the Particular Specifications. Horizontal flanges or other structural surfaces which may collect condensation below the covers shall be avoided. Where supporting members are proposed to be structural steel, tee and angle sections shall be used in preference to channel or I-beam sections. Where horizontal structural surfaces cannot be avoided, suitable provision shall be made to avoid pooling, and for drainage of condensation.
 - (iv) Structural Design shall be carried out in accordance with referenced Standards and the appropriate materials standards.
 - (v) A Design Certificate shall be issued by a P.E. for the structural performance and sufficiency of the odour containment covers.
 - (vi) A copy of all calculations shall be provided to the S.O..
- (b) Design Inputs
 - (i) The Contractor shall be required to complete the structural and mechanical design of the odour containment covers in accordance with relevant standards and appropriate material standards.
 - (ii) Odour cover components shall be designed in accordance with the relevant material Standards or Industry Guidelines. Current editions (at the time of tender) of each document shall be used for the design.

- (iii) The maximum deflection of any component under the loads listed herein shall not exceed 1/240 of the span of that component. To avoid ponding, dead load deflection shall not exceed the rise of any component.
- (iv) The Contractor shall submit a summary of calculated deflections as part of the design submission.
- (v) The design shall make all necessary allowances for thermal expansion and contraction
- (vi) Further details on design inputs for the odour containment covers can be found in the Particular Specifications.

G4.73.11 Product Delivery, Storage and Handling

- (a) General
 - (i) The covers and associated components shall be delivered to site. The cost of shipping and delivery of said equipment shall be deemed to be included in the Contract Sum.
 - (ii) The S.O. may elect to inspect the cover components to be supplied by the Contractor prior to packaging and delivery. The S.O. will give a minimum of TWO (2) working weeks' notice to the Contractor of any intended inspections.
- (b) Delivery, Storage and Handling Requirements
 - (i) The Contractor shall forward a packing list to the S.O. a minimum of TEN (10) working days prior to delivery of the covers, which details the number, size and weight of each item (box, crate, etc.), and the contents of each item.
 - (ii) All components delivered to site shall be packed to protect against damage during transport and unloading.
 - (iii) Odour cover components stored onsite shall be sealed and protected; suitable for outside storage for duration of up to NINETY (90) days.
 - (iv) Appropriate measures shall be taken to prevent the entrance of moisture or water to packaged components during shipment and storage onsite.
 - (v) Each item of the delivered odour cover components shall be suitably labelled with the project name, reference number, the Contractor's name and contact details, and shall be accompanied with the packing list, and a delivery docket.

G4.73.12 Testing and Commissioning

- (a) Factory Acceptance Test (FAT)
 - (i) General
 - 1) All relevant FAT test procedure and test schedules shall be submitted to the S.O. for approval minimum of TWO (2) weeks before the commencement of FAT.

- 2) The equipment will be subjected to a thorough visual inspection to confirm that:
 - All components / systems comply with the relevant specifications / drawings;
 - All factory assembled components / systems are correctly assembled / installed in accordance with the relevant PID and GA drawings; and
 - All warning signs and safety labels are fitted in accordance with the relevant drawings.
 - 3) Should the supplied covers fail any test, the Contractor shall take any actions deemed necessary by himself or the S.O. to correct the failure and repeat the test.
 - 4) Prior to, during and subsequent to any test, the Contractor shall adhere to all required safety protocols and standards; including but not limited to method statements and risk assessments.
 - 5) The S.O. reserves the right for its representatives to be present for any of the required testing if deemed necessary.
- (b) Factory Witness Testing (Where Required)
- (i) All materials, components, assemblies and fabrications used in the Works are subject to factory inspection by the S.O. and the Contractor shall advise at least FOUR weeks prior to date when such items are ready for inspection
 - (ii) The Supplier's FAT schedules shall be available at the witness test, having been previously completed by the Supplier and signed by him to confirm that FAT has been satisfactorily carried out.
 - (iii) The S.O. shall give the Contractor TWO working weeks' notice of their intention to inspect.
 - (iv) The inspection and testing activities to be repeated shall be at the discretion of the S.O..
- (c) Site Acceptance Testing (SAT)
- (i) General
 - 1) All relevant SAT test procedures and test schedules shall be submitted to the S.O. for approval, a minimum of TWO weeks before the commencement of SAT;
 - 2) Installation of odour containment covers on a "live" tank or channel shall follow the Board's relevant Standard Operating Procedures (SOPs). The Contractor shall submit all plans to the S.O. for approval;
 - 3) The Contractor shall make allowance for any cut-overs on "live" tanks or channels as required in his programme for the Works. Detailed method statements for all installation works and procedures shall be submitted to the S.O. for review and approval;

- 4) After the covers have been installed, acceptance checks will be performed to confirm the satisfactory installation and construction completeness, with the Manufacturer's representative in attendance; and
- 5) The S.O.'s representatives may be present for any testing.

(ii) Commissioning Tests

- 1) Commissioning tests shall include, but not be limited to:
 - Checking the horizontal and vertical alignment of all covers;
 - Checking the operation of safety barriers;
 - Check operation of failsafe automatic opening covers;
 - Checking the operation of rainwater disposal system;
 - Checking the operation of hatches;
 - Examining protective coatings and repair where necessary;
 - Examining gasket/sealing and repair where necessary; and
 - Checking the location and completeness of ductwork connections.
- 2) All necessary corrections and adjustments shall be made by the Contractor. A written report shall be provided at the completion to confirm the covers have been properly installed and operate satisfactorily;
- 3) On-site tests during commissioning shall be carried out to clearly demonstrate and record that the covers meet the specified performances and have been successfully commissioned as a complete and integrated installation; and
- 4) The Contractor shall supply all necessary testing equipment, measuring instruments and the appropriately skilled labour required for conducting the tests. Should testing show the performance to be less than specified, then all necessary changes shall be made to the covers for the performance criteria to be satisfied. All associated costs shall be borne by the Contractor.

(iii) Deflection Tests

- 1) Deflection of cover assemblies shall be tested and verified after installation for conformance with the deflection limits specified herein. An approved testing laboratory shall be procured to certify and report results. A 200 kg load shall be placed as directed by the S.O. and the maximum deflection measured.

(d) Performance Testing

(i) General

- 1) Performance testing of covers is a two-step process;

- 2) The first step is smoke testing to identify major leaks in ductwork and covers. This is performed under slight positive pressure using a fan and a smoke machine. This is an effective method for locating missing gaskets, ill-fitting covers and other defects;
- 3) Once the smoke testing is complete and defects rectified, infiltration rates from leaks can be determined. This is done by measuring the airflow through intake and outlet ductwork with the balance attributed to leaks. The negative pressure under the covers shall also be measured at specified locations to endure that there are no dead zones under the covers;
- 4) A written report shall be provided at the completion of testing to confirm the covers have been properly installed and operate satisfactorily;
- 5) Should testing show the performance to be less than specified, then all necessary changes shall be made to the covers installation for the performance criteria to be satisfied; and
- 6) Following installation of the equipment at site, any components/systems that have not been previously inspected during FAT or have been newly assembled/installed at site shall be subjected to a thorough visual inspection to confirm that they have been correctly assembled/installed in accordance with the relevant PID/GA drawings.

(ii) Smoke Testing

- 1) Smoke testing should be performed to prove the ability of the covers to maintain a negative pressure beneath them, without excessive leakage from joints.
- 2) Smoke Testing Protocol:
 - Measure the volume of the void between the top water level and the underside of the covers in m³;
 - Select a suitably sized smoke generator, to produce the relevant volume (m³) of smoke time not exceeding 20 minutes;
 - The smoke should be non-toxic and water based;
 - Prior to starting the test:
 - Stop the odour control equipment fans;
 - Isolate the covers from the odour control equipment; and
 - Close all cover inlet ports and hatches;
 - Connect the smoke generator to a suitable spigot connection on the covers, via a 75 mm diameter flexible hose;
 - Start the smoke generator and when the fill time has elapsed, inspect the covers for signs of smoke leakage and rectify any areas that are leaking;

- Re-connect the covers to the odour control equipment and start the odour control equipment fans; and
 - Observe the areas that were leaking and, if smoke is still visible, take further remedial action to seal any leaks.
- (iii) Infiltration Rate Testing
- 1) Infiltration Rate Testing is the second step of the performance testing. Once the smoke testing is complete and defects rectified, infiltration rates from leaks can be determined.
 - 2) Infiltration Rate Testing Protocol
 - Measure the airflow through intake and outlet ductwork, and attribute the balance to leaks;
 - The negative pressure under the covers shall also be measured at specified locations to ensure that there are no dead zones under the covers; and
 - Air pressure gauges installed through the covers shall read a negative pressure of at least 20 Pa. g at the design air flowrate to the Odour Control Equipment and covers air intakes set at design set points.
- (iv) Cover Leakage Testing (Where Required)
- 1) Leakage testing shall be performed to prove the ability of the covers to maintain a negative pressure beneath the cover and without excessive leakage from joints; and
 - 2) The Contractor shall propose the leak test procedure for S.O.'s approval.
- (v) Process Performance Testing
- 1) Following Completion Testing, process performance tests shall be conducted to prove that the equipment performs in accordance with the long-term performance requirements; and
 - 2) The test details, number of test periods and duration of each test period shall be fully in accordance to the Testing and commissioning and shall be at the discretion of the S.O..
- (vi) Further Tests (Where Required)
- 1) Details of any further performance tests to be performed on the treatment unit shall be as specified in particular specification and as S.O. instruction without incur any additional cost.
- (vii) Reporting of Results
- 1) All testing results shall be made available to the S.O. within a period of TEN working days or unless otherwise specified in the Particular Specification;
 - 2) On completion of testing, all results shall be collated and presented as a single electronic copy with all supporting calculations provided; and

- 3) If any tests are failed, then the report shall identify the cause of failure and detail any remedial actions which have been or will be taken to achieve compliance with the requirements of this Specification.

G4.73.13 Operation and Maintenance Manuals

- (a) Submit Operation and Maintenance manuals at least THREE weeks before the date of delivery of the equipment to the relevant project site.
- (b) The information to be submitted shall comply with the requirements specified in the General Specifications G1.2 Information Management.

G4.73.14 Spare Parts and Special Tools

- (a) The Contractor shall nominate as a minimum, details of all replaceable spare parts for all routine and preventative maintenance, and repair procedures, including a recommended list of spare parts for operating periods of 12 months, 2 years, and 5 years. This list shall include:
 - (i) Description, including full specification details;
 - (ii) Serial number, part number, make, and model;
 - (iii) Manufacturer, Contractor's and Local Agent Details; and
 - (iv) Origin of spare part and delivery lead time.
- (b) The Contractor shall also nominate a full listing of any special tools or equipment that might be required to carry out routine or preventative maintenance, repair procedures, equipment adjustments or the replacement of any spare parts

END OF SECTION